Department of Computer Science and Engineering
Academic Year 2021 - 2022 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. Computer Science and Engineering
Course Code & Title: CS8451 & Design and Analysis of Algorithms
Name of the Faculty member (s): Ms.P.Jothi Thilaga, AP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit III / Dynamic Programming and Greedy Technique
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO6 & TLO7
- **Activity Chosen:** STAD
- **Justification:**

  Dynamic Programming and Greedy Technique are the most important algorithmic problem solving techniques in design and analysis of algorithms. This activity makes the students to get a sound knowledge to solve the problems using these two different techniques. Students think individually and solve the problems by discussing with their team members can improve their knowledge in the particular topic.

- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**
  - Initially the groups were formed based on the student's willingness. 13 teams were formed with a group of 4 or 5 students. Team members selected a team name based on their interest.
  - Each team was assigned with a different set of question. They have to solve the given set of questions by discussing with their team members. 15 minutes were allocated to solve the given set of question.
  - Finally, each team discusses the methods and steps to solve the allotted problems within the given time to the entire class.
  - Based the rubrics designed, each teams were graded. The rubric for this activity is given in the Table 1.1.

Table 1.1: Rubric for STAD Activity

Evaluation criteria	Excellent 5.0 to 4.0 pts	Good 3.0 to 2.0 pts	Satisfaction 2.0 to 0 pts	Marks
Knowledge in the given topic	Every Student in the team can able to solve the problems individually.	Student in the team can able to solve the problems individually, but needs some additional support.	Student in the team feel some difficulties in solving the problems individually.	**5**
Time Management	Answer all the questions correctly within the allotted time	Answer the questions partially within the allotted time	Answer the questions beyond the allotted time	**5**
Group Coordination	Each individual performed excellently and coordinated well with the team members	Each individual performed good and coordinated well with the team members	Each individual performed good and coordination among the team members is lacking	**5**
Group Presentation	The presentation was divided equally to individual in a group. The individual performed excellently.	The presentation was somewhat divided to an individual in a group. An individual was somewhat performed well	The presentation was not equally divided into an individual in a group. An individual was not performed well	**5**
			Total	**20**

The individual score will be calculated based on the above rubrics such as Knowledge in the given topic, Time Management, Group Coordination, Group Presentation. The final team score was calculated by taking the average of the individual scores in each team and it is listed in the Table 1.2.

Table 1.2: Team wise Scores

Sl. No.	Reg. No.	Name of the Student	Group Name	Marks (20)
1.	953620104002	ABINAYA M	Sunshine	18
2.	953620104028	LAKSHMI PRIYA G		
3.	953620104053	SRINIDHI R		
4.	953620104054	S SHANMUGAPRIYA		

5.	953620104005	BHUVANESHWARI S	Null	19
6.	953620104010	GAYATHRI DEVI S		
7.	953620104018	JELENA JOPHIN J		
8.	953620104047	SHAKTHI S		
9.	953620104048	SHRI DHEIVA R		
10.	953620104006	BOWYAA L	Sectum Sempra	18
11.	953620104022	KAMINI K		
12.	953620104024	KARUNYA V		
13.	953620104025	KEERTHANA V		
14.	953620104034	PAVITHRA K		
15.	953620104007	DHARANI RAJ V	Zankar	18
16.	953620104046	SHAKTHEE SARAVANA M		
17.	953620104050	SIVAKUMAR M		
18.	953620104305	SHANKAR S K		
19.	953620104008	DHIRAJ N	Nullvoid	19
20.	953620104011	GNANAGURU VIGNESH J		
21.	953620104302	ILAYA BHARATHI S		
22.	953620104306	SUBRAMANI LOKESH		
23.	953620104009	DIVYA M K	Infinity	20
24.	953620104035	POOJALAKSHMI T		
25.	953620104036	P PREETHA		
26.	953620104052	SOWMIYA V		
27.	953620104060	VISHNUPRIYA S		
28.	953620104013	GOWTHAM KARTHIKEYAN M	Matrix	19
29.	953620104021	JOTHI LINGAPANDIAN M		
30.	953620104049	SIVAGURUNATHAN G		
31.	953620104056	SUKUMARAN G		
32.	953620104014	HARINISRIVALLI V	5 Minds	20
33.	953620104038	RAGAVI V G		
34.	953620104040	RAJALAKSHMI R U		
35.	953620104042	SARAVANA JOTHI G		
36.	953620104058	VENESHA L		
37.	953620104015	HARSHA R	Greedy	20
38.	953620104031	MUTHU KUMAR K		
39.	953620104033	PARANTHAMAN J		
40.	953620104045	SATHISHKUMAR K		
41.	953620104017	JEGADESH B	Alambal	18
42.	953620104019	JOE ALAN R S		
43.	953620104304	SATHYA NARAYANAN B		

44.	953620104023	KARTHIKEYAN B	Empty	18
45.	953620104029	MUGESH M		
46.	953620104041	RAJASEKARAN T		
47.	953620104044	SATHISH KANNAN M		
48.	953620104059	VINO A		
49.	953620104027	KUMAR M	Dharsan	20
50.	953620104030	MUTHU BRIJESH R		
51.	953620104051	SIVAKUMAR P		
52.	953620104303	RAAMAKRISHNAN V R		
53.	953620104301	BALAMURUGAN K		
54.	953620104032	NISHASHREE M	Rush	20
55.	953620104039	RAJA DHARSHINI S		
56.	953620104043	SARUMATHI M		
57.	953620104055	SUBHIKSHA G		

- **CO – PO / PSO mapping:**

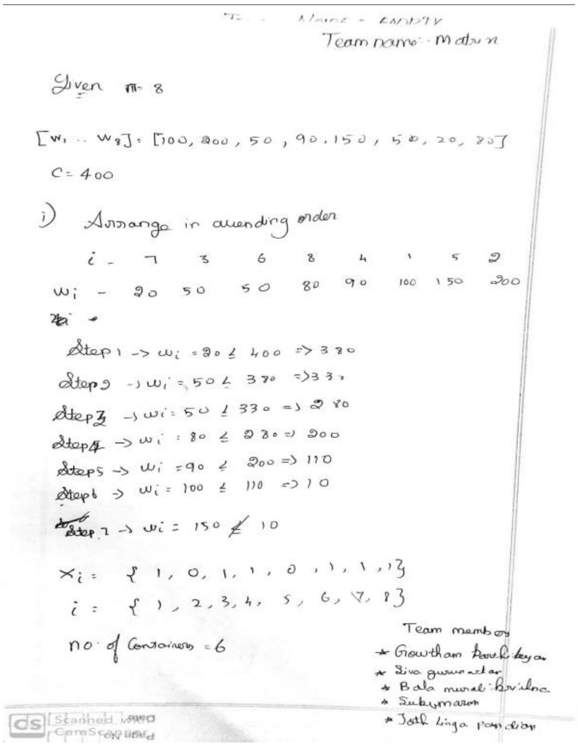
CO	PO1	PO2	PO3	PO4	PO10	PSO1	PSO2	PSO3
CO1	3	1	1	1	2	2	2	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**
	3	1	1	1
Justification for correlation	Different algorithm design techniques are used for analysis of algorithm.	Different algorithm design techniques used in analysis of engineering problems	To identify solution for complex problems using this algorithm design technique	To identify solution for complex problems using algorithms.
	PO10	**PSO1**	**PSO2**	**PSO3**
	2	2	2	1
	To present solution for complex problem solutions with specific algorithms	Able to use these algorithm design techniques in Information Technology	Able to use these algorithms in analyzing and developing reliable IT Solutions	Algorithm design techniques are helpful in solving real world problems in Industry and Research.

- **Images / Screenshot of the practice:**



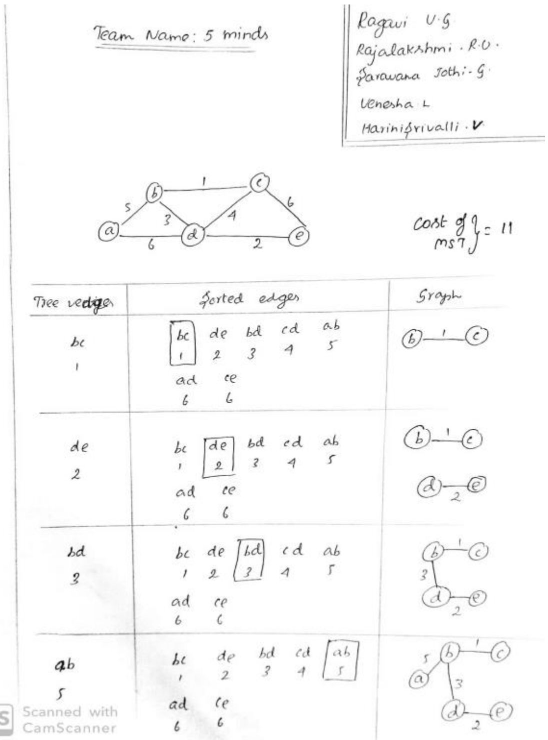


Figure 1: Screenshots of STAD activity

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - All the students actively participated and enjoyed among their group members once they were able to solved the problem correctly.
    - Students explored their knowledge on the dynamic programming and greedy technique by solving the problem.
    - It makes the students to gain well knowledge on how to apply these techniques on different problems.
  - ***Benefit of the practice:***
    - Students were actively participated in this activity.

- From this activity, each student can refresh the dynamic Programming and greedy technique concept and understood the topic clearly while the information is shared among them.

❖ ***Challenges faced in implementation:***
- Due to time constraints, only few group of students were made to share the concepts by a random call.
- Slow learners were felt difficult to share the concepts in-front of others.

References:

- https://www.ritrjpm.ac.in/images/computer-science/CS8601-Mobile%20Computing-STAD.pdf
- http://cw.routledge.com/textbooks/9780415965286/studentresources_chapter8cooperative.asp
- https://iopscience.iop.org/article/10.1088/1757-899X/336/1/012027/pdf

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